

Finite-Time Feedback Control with Social Context

Single-affinity control with arbitrary integer $q \geq 1$

31 August 2026

Abstract

We extend the finite-time single-affinity feedback-control theory from the special controlled layer $q = 1$ to arbitrary integer social group size $q \geq 1$. At a controlled microscopic update, one social slot is occupied by the controller while $q - 1$ ordinary peers remain. Their composition is summarized by a residual-peer hypergeometric law and enters the controlled transition kernel only through a symmetric edge activity $G_q(n)$. The controller retains one measurable directional affinity h . This separation is enough to prove that the forward/reverse transition ratio, weighted reversibility, and the complete finite-time feedback path identity are unchanged in form:

$$\Delta S_{\text{sys}} + hJ_{c,q} + I(n; Y) = \Sigma_q = D_{\text{KL}}(P_F^{(q)} \| P_R^{(q)}) \geq 0.$$

Thus q changes controlled traffic, susceptibility, action-to-population information, current, final occupancy and irreversibility, but does not by itself introduce a new thermodynamic force. The exact generalized susceptibility is obtained from the finite matrix kernel rather than forced into a closed form. We preserve the bounded information-response efficiency and the finite-time thermodynamic control efficiency. A full numerical parity suite confirms that the implementation reduces to the previous $q = 1$ theory to machine precision. For the theoretical figures we use one explicitly illustrative smooth “corroboration” kinetic closure; the theorems do not depend on that choice.

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1 Problem and one-cycle architecture

Fix a target opinion Z in a population of N agents. Immediately before feedback cycle k , let

$$n_k \in \{0, 1, \dots, N\}, \quad x_k = \frac{n_k}{N},$$

where n_k is the number of agents supporting Z . The controller samples the population, chooses an action, and then the isolated controlled layer updates the population:

$$n_k \longrightarrow Y_k \longrightarrow U_k \longrightarrow n_{k+1}.$$

The finite-time population ensemble is

$$p_k(n) = P(n_k = n),$$

and no stationarity assumption is made. In particular, p_{k+1} may differ substantially from p_k .

The social group size q is distinct from the sensor size q_c and the actuation budget b . Operationally, a focal agent normally receives q social slots. At a controlled position exactly one slot is replaced by the controller, so the controlled update contains

controller + $(q - 1)$ ordinary peers.

For $q = 1$, the controlled focal sees the controller alone. For $q > 1$, the intervention is embedded in residual social context.

The conceptual chain is

sensing/policy $\rightarrow K_q \rightarrow \chi_q \rightarrow T_{\pi,q}(n) \rightarrow J_{c,q} \rightarrow \Sigma_q \rightarrow \eta_{\text{th},q}.$

2 Sensing and feedback decision

The sensing and policy pieces are unchanged. The controller samples q_c agents without replacement. If $Y_k = y$ of them support the target, then

$$S(y | n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}.$$

The stochastic advocacy policy is

$$\pi(1 | y) = \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right],$$

and the sensor-averaged action probability at population state n is

$$a_n = P(U_k = 1 | n_k = n) = \sum_y S(y | n) \pi(1 | y).$$

The social group size q is deliberately absent from these equations: it enters only in the actuation/social-context layer.

3 Residual-peer context for arbitrary q

Consider the neighboring population edge $n \leftrightarrow n + 1$. Let

$$J \in \{0, 1, \dots, q - 1\}$$

be the number of target-supporting agents among the $q - 1$ residual ordinary peers at a controlled opportunity.

For an upward transition $n \rightarrow n + 1$, the focal agent is non-target. After excluding the focal, the peer pool contains n target agents and $N - n - 1$ non-target agents. Therefore

$$H_q(j \mid n) = \frac{\binom{n}{j} \binom{N-n-1}{q-1-j}}{\binom{N-1}{q-1}}.$$

For the reverse transition $n + 1 \rightarrow n$, the focal agent is target. Excluding it again leaves exactly n target and $N - n - 1$ non-target agents. Hence the same $H_q(j \mid n)$ describes both directions of the same edge. This edge-pair identity is the key structural observation.

3.1 Context-dependent kinetic compliance

Introduce a bounded nonnegative kinetic compliance

$$\gamma_q(j) \in [0, 1].$$

It is an activity/traffic factor, not an additional directional affinity. The edge-averaged activity is

$$G_q(n) = \sum_{j=0}^{q-1} H_q(j \mid n) \gamma_q(j), \quad n = 0, \dots, N - 1.$$

At $q = 1$ the residual peer set is empty, and we require

$$\gamma_1(0) = \gamma, \quad G_1(n) = \gamma.$$

The implementation provides three main modes. **constant** sets $\gamma_q(j) = \gamma$ and is a null model in which q has no kinetic effect. **context_table** accepts an empirically calibrated vector $\gamma_q(0), \dots, \gamma_q(q - 1)$. For theory-only figures we use an explicitly illustrative smooth closure,

$$\gamma_q^{\text{can}}(j) = \gamma \frac{j + 1}{q}.$$

The $+1$ counts the controller itself, which always advocates the target. Under this closure,

$$\mathbb{E}[J \mid n] = (q - 1) \frac{n}{N - 1},$$

so

$$G_q^{\text{can}}(n) = \frac{\gamma}{q} \left[1 + (q - 1) \frac{n}{N - 1} \right].$$

This closure is useful because it is smooth, bounded, monotone, and exactly recovers $q = 1$. It is not asserted to be the behavioral law of the LLM population.

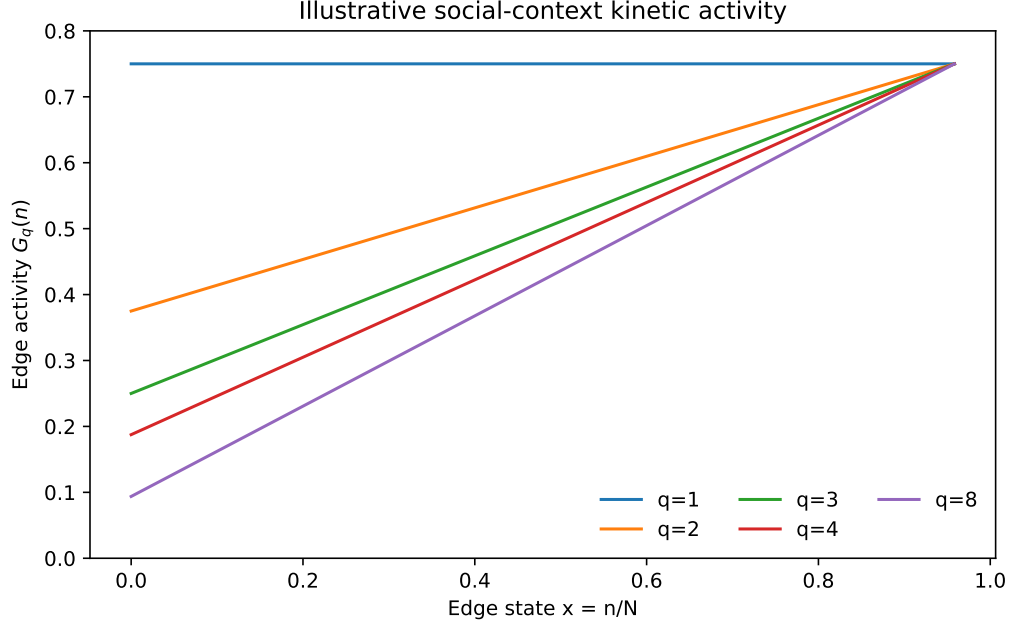


Figure 1: Illustrative edge activity $G_q(n)$ for the canonical corroboration gate with $N = 24$ and $\gamma = 0.75$. Increasing q suppresses controlled activity most strongly at low target support while leaving high-support edges comparatively accessible. This is a kinetic effect only; the edge affinity remains h .

4 Generalized controlled kernel

Let

$$p_h = \sigma(h).$$

The generalized one-opportunity controlled birth-death kernel is

$$K_q(n+1 | n) = \frac{N-n}{N} G_q(n) p_h, \quad n = 0, \dots, N-1,$$

$$K_q(n-1 | n) = \frac{n}{N} G_q(n-1) (1-p_h), \quad n = 1, \dots, N,$$

and

$$K_q(n | n) = 1 - K_q(n+1 | n) - K_q(n-1 | n).$$

Because $0 \leq G_q \leq 1$, every row is stochastic. The action-conditioned cycle kernels preserve the semantics of the revised $q = 1$ theory:

$$Q_0 = I, \quad Q_1^{(q)} = K_q^b.$$

Ordinary peer-only diffusion is not inserted into Q_0 ; the reference isolates the controlled channel.

At $q = 1$, $G_1 = \gamma$ and therefore

$$K_1(n+1 | n) = \gamma \frac{N-n}{N} \sigma(h), \quad K_1(n-1 | n) = \gamma \frac{n}{N} \sigma(-h),$$

which is exactly the previous finite-compliance kernel.

5 Exact generalized susceptibility

For general q , state-dependent edge activity prevents a useful universal closed form after b controlled opportunities. The exact finite- N susceptibility is therefore defined directly from the matrix kernel. With $x_m = m/N$,

$$\chi_q(n/N) = \mathbb{E}[x' \mid U = 1, n, q] - \mathbb{E}[x' \mid U = 0, n, q] = \sum_{m=0}^N \left(\frac{m-n}{N} \right) Q_1^{(q)}(m \mid n).$$

Equivalently,

$$\chi_q = (Q_1^{(q)} x) - x.$$

This definition is exact for every integer $q \geq 1$ and every admissible kinetic-context profile.

The one-opportunity target-count drift is

$$d_q(n) = \frac{N-n}{N} G_q(n) p_h - \frac{n}{N} G_q(n-1) (1-p_h).$$

For $q = 1$ the matrix definition collapses to the previous analytic expression

$$\chi_1(x) = [\sigma(h) - x] \left[1 - \left(1 - \frac{\gamma}{N} \right)^b \right].$$

Thus $x^* = \sigma(h)$ is the exact zero-response point at $q = 1$. For $q > 1$ it should be computed from the exact χ_q curve rather than assumed: state-dependent kinetics can shift the finite-step zero even though the edge-level affinity remains unchanged.

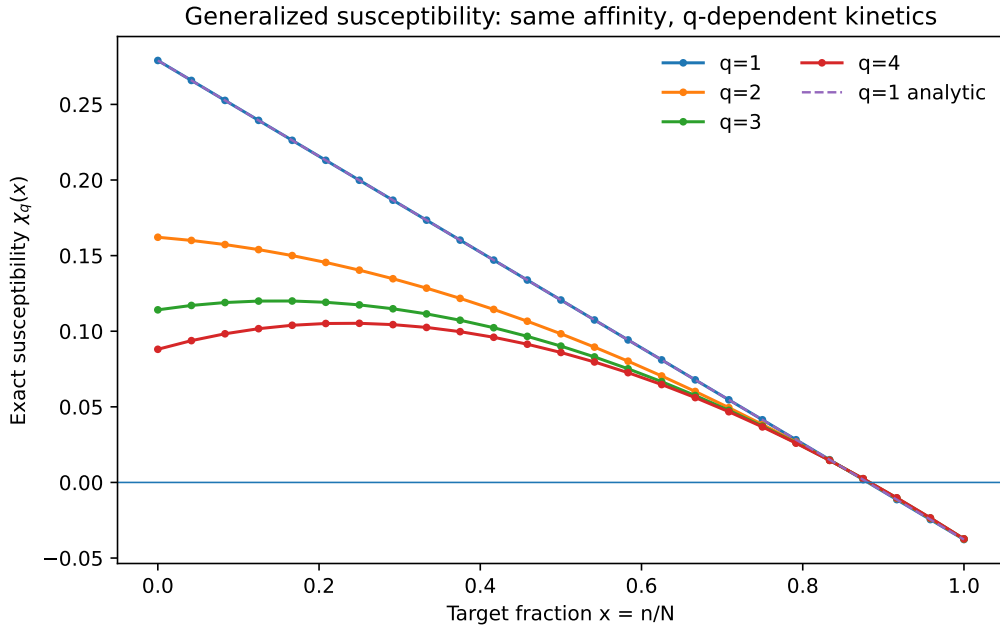


Figure 2: Exact generalized susceptibility at $N = 24$, $h = 2$, $b = 12$, $\gamma = 0.75$ using the illustrative canonical group gate. The dashed curve is the old analytic $q = 1$ result and overlaps the generalized $q = 1$ calculation to numerical precision. Additional residual peers suppress low-support responsiveness and can bend the response profile without changing the directional affinity.

A useful consequence is that an “interior susceptibility window” is not a theorem of $q > 1$ by itself. It depends on how social context changes kinetic accessibility. Figure 3 contrasts three admissible closures while all retain the same h and the same thermodynamic transition-ratio structure.

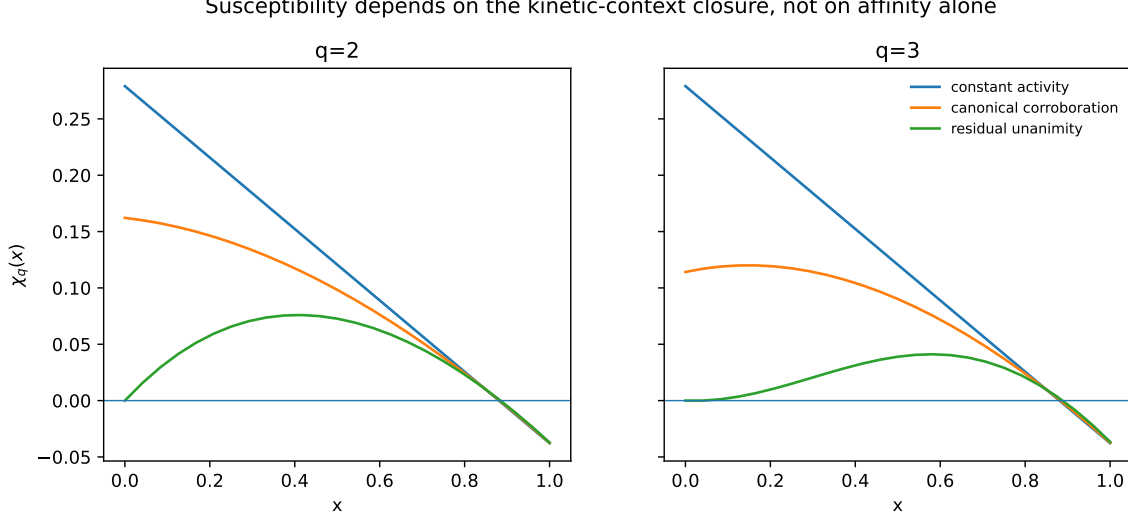


Figure 3: Closure sensitivity of χ_q . Constant activity leaves the response identical to the $q = 1$ kernel; the smooth canonical corroboration gate suppresses low-support response; a residual-unanimity gate produces the stronger interior window familiar from a q -voter-style mechanism. The general thermodynamic theorem below holds for all three because the activity remains edge-symmetric.

6 Exact state-local action-to-population information

At fixed state n , the closed-loop next-state kernel is

$$Q_{\pi}^{(q)}(m | n) = (1 - a_n)Q_0(m | n) + a_n Q_1^{(q)}(m | n).$$

The exact state-local action information is

$$T_{\pi,q}(n) = I(U_k; n_{k+1} | n_k = n, q) = \sum_{u \in \{0,1\}} P(u | n) \sum_m Q_u^{(q)}(m | n) \log_2 \frac{Q_u^{(q)}(m | n)}{Q_{\pi}^{(q)}(m | n)}.$$

It is the weighted Jensen-Shannon divergence between the NoOp and advocacy next-state distributions. Action variability and kernel separation are both required. Therefore strong open-loop actuation can coexist with small $T_{\pi,q}$ if the policy is nearly deterministic, while large $T_{\pi,q}$ need not correspond to a large signed mean response.

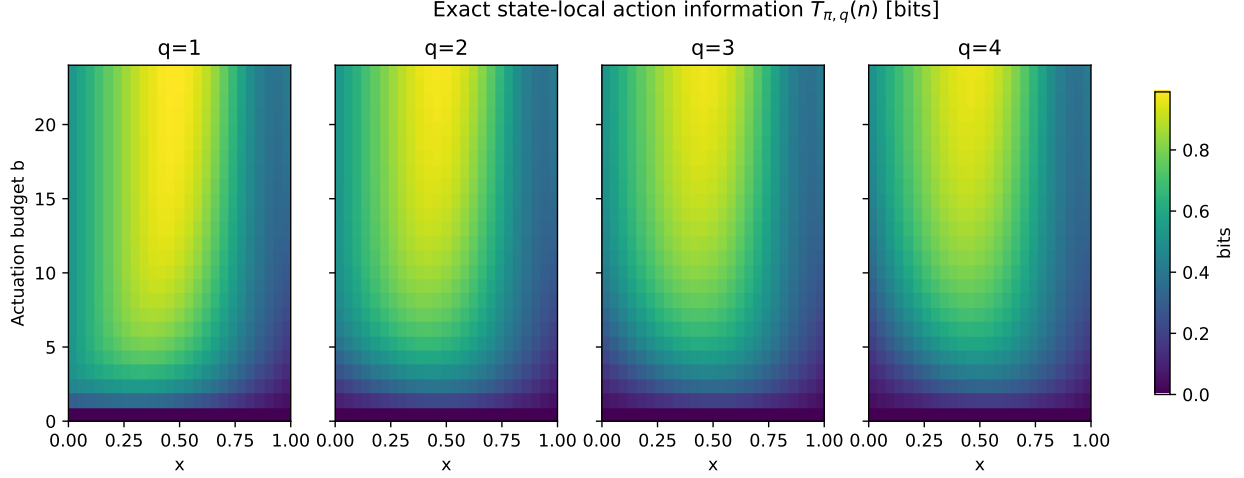


Figure 4: Exact $T_{\pi,q}(n)$ landscapes for $N = 24$, $h = 2$, $q_c = 12$, $\beta = 4$, $\theta = 1/2$, $\gamma = 1$ and the canonical group gate. Increasing q changes the kinetic separation of the two action-conditioned kernels and redistributes the information landscape while leaving the policy itself unchanged.

6.1 Bounded information-response conversion

Pinsker's inequality and the bounded observable $m/N \in [0, 1]$ give

$$T_{\pi,q}(n) \geq \frac{2a_n(1-a_n)}{\ln 2} \chi_q(n/N)^2.$$

Whenever $T_{\pi,q}(n) > 0$, define

$$\eta_{\text{IR},q}(n) = \frac{2a_n(1-a_n)\chi_q(n/N)^2}{(\ln 2)T_{\pi,q}(n)} \leq 1.$$

This is an information-response conversion fraction, not the thermodynamic efficiency derived later.

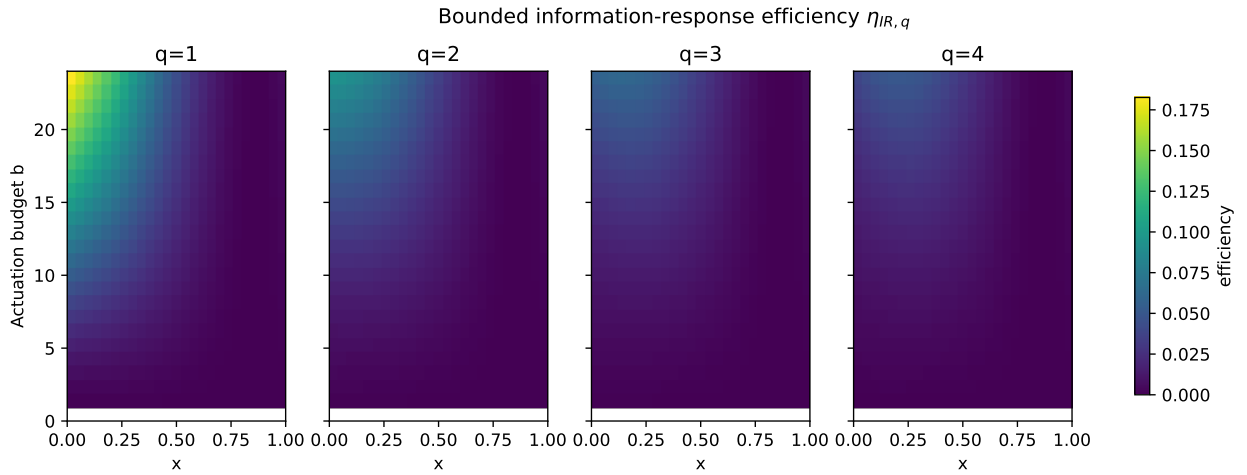


Figure 5: Exact state-local information-response efficiency for the same baseline as the preceding figure. In the illustrative closure, increasing q generally lowers the fraction of action-visible distributional change expressed as coherent mean target motion.

7 Parameter sweeps and explicit q dependence

The previous report emphasized that the local information landscape depends separately on affinity, sensing, policy gain and threshold. That structure remains. Figures 6–9 reproduce the same arithmetic state-average summary for $q = 1$ and one representative $q = 2$ slice. These averages do not assume any population ensemble.

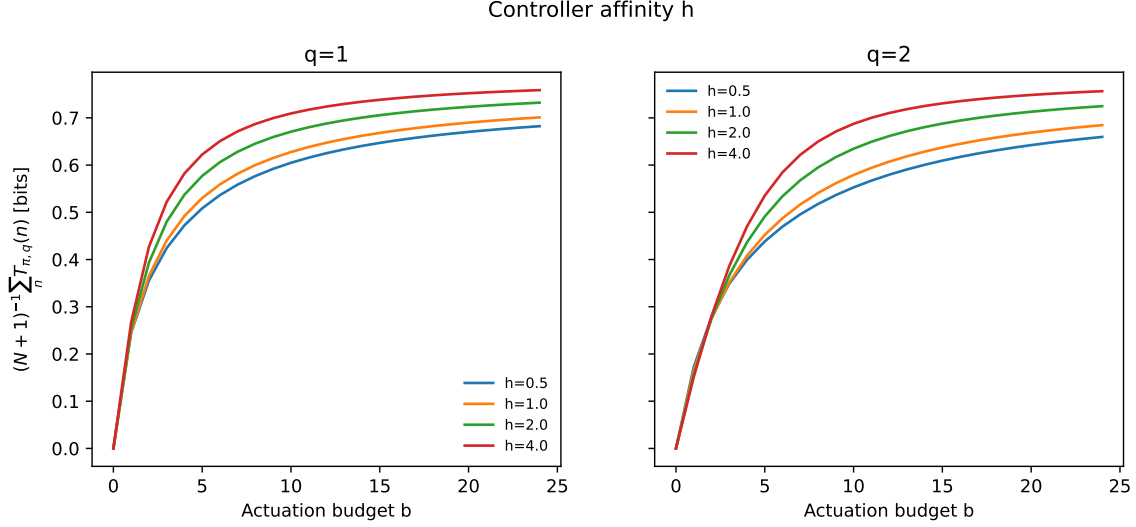


Figure 6: Arithmetic state-average $T_{\pi,q}$ while varying the controller affinity h , shown for $q = 1$ and $q = 2$.

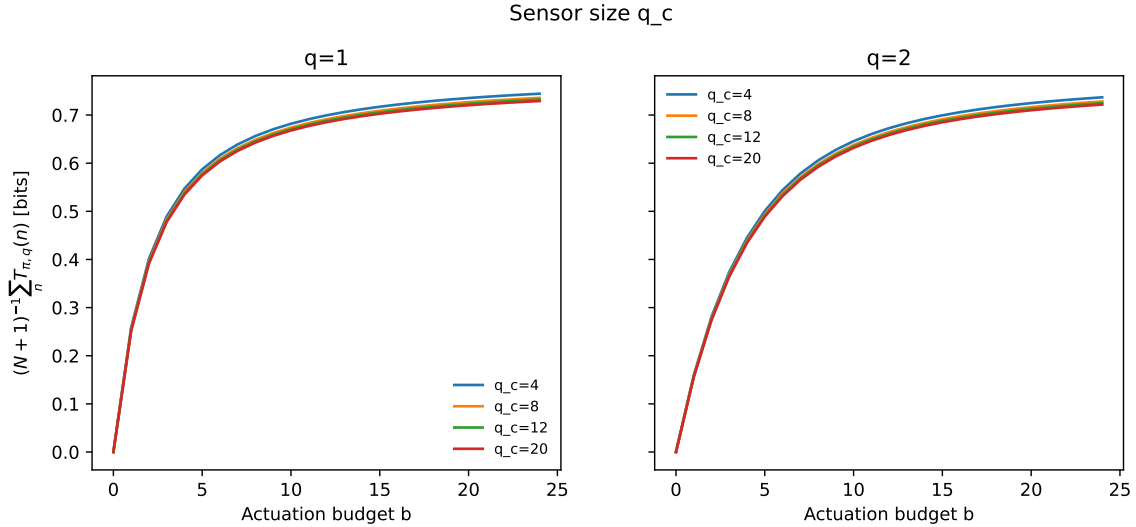


Figure 7: Arithmetic state-average $T_{\pi,q}$ while varying the sensing sample size q_c . Sensing changes a_n and hence the action-information ceiling, not the microscopic controlled kernel.

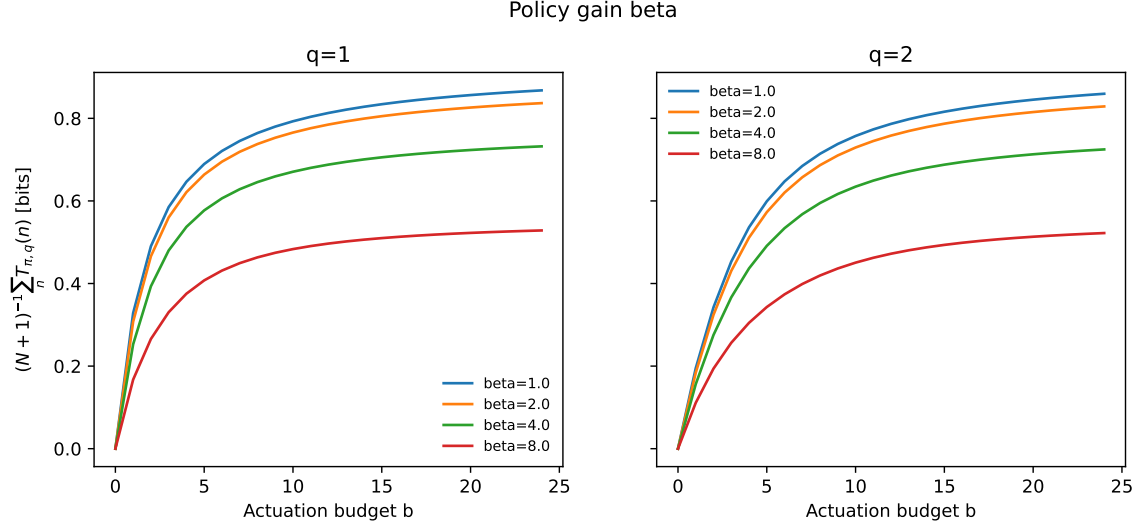


Figure 8: Arithmetic state-average $T_{\pi,q}$ while varying policy gain β . A more switch-like policy can sharpen the informative region while reducing information where the action saturates.

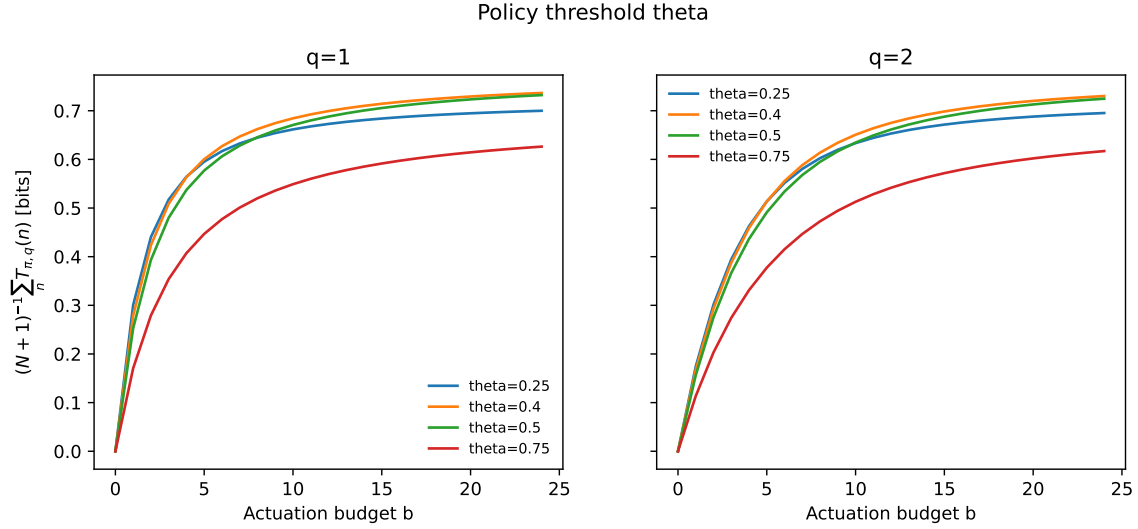


Figure 9: Arithmetic state-average $T_{\pi,q}$ while varying policy threshold θ .

The new independent sweep is q itself:

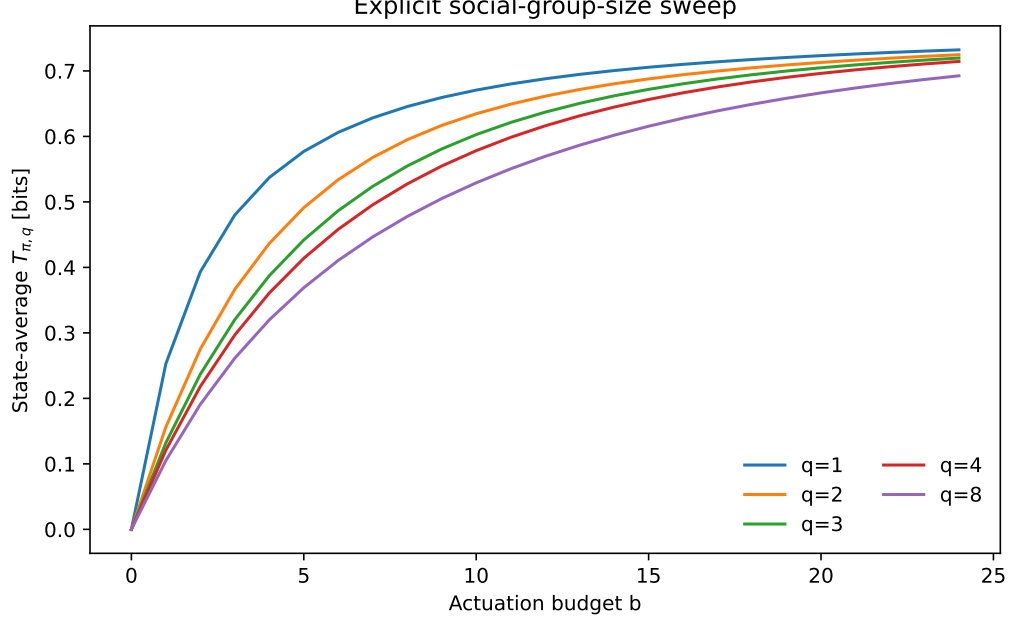


Figure 10: Explicit social-group-size sweep at $N = 24$, $h = 2$, $q_c = 12$, $\beta = 4$, $\theta = 1/2$, $\gamma = 1$. Under the illustrative canonical gate, increasing q lowers the arithmetic state-average action information because controlled transitions become kinetically less accessible over much of state space.

8 Generalized controlled current

For an ensemble p_k , the exact mean target-count current attributable to the controlled layer is

$$J_{c,q,k} = N \sum_n p_k(n) a_n \chi_q(n/N).$$

Positive $J_{c,q}$ denotes net movement toward the controller target; negative values denote net movement away. The $q = 1$ expression reduces exactly to the closed form in the previous report.

9 Central transition-ratio theorem for arbitrary q

For each neighboring edge, the generalized kernel gives

$$K_q(n+1 | n) = \frac{N-n}{N} G_q(n) \sigma(h),$$

while the reverse transition is

$$K_q(n | n+1) = \frac{n+1}{N} G_q(n) \sigma(-h).$$

The same edge activity appears in both directions and therefore cancels:

$$\frac{K_q(n+1 | n)}{K_q(n | n+1)} = \frac{N-n}{n+1} \frac{\sigma(h)}{\sigma(-h)} = \frac{N-n}{n+1} e^h.$$

Define the population mixing entropy

$$S_{\text{mix}}(n) = \ln \binom{N}{n}.$$

Since

$$S_{\text{mix}}(n+1) - S_{\text{mix}}(n) = \ln \frac{N-n}{n+1},$$

we obtain the edge-level local-detailed-balance relation

$$\ln \frac{K_q(n+1 | n)}{K_q(n | n+1)} = S_{\text{mix}}(n+1) - S_{\text{mix}}(n) + h.$$

This identity is independent of q and of the particular admissible $\gamma_q(j)$ profile.

Now define

$$w_h(n) = \binom{N}{n} e^{hn}.$$

The edge relation is equivalent to detailed balance with respect to w_h ,

$$w_h(n) K_q(m | n) = w_h(m) K_q(n | m).$$

Detailed balance is preserved by powers, so

$$w_h(n) Q_1^{(q)}(m | n) = w_h(m) Q_1^{(q)}(n | m),$$

and hence

$$\ln \frac{Q_1^{(q)}(m | n)}{Q_1^{(q)}(n | m)} = S_{\text{mix}}(m) - S_{\text{mix}}(n) + h(m - n).$$

This is the central theoretical payoff: residual social context changes traffic while the thermodynamic directional term remains a single affinity h .

10 Forward and reverse finite-time feedback paths

For one feedback cycle,

$$P_F^{(k,q)}(n, y, u, m) = p_k(n) S(y | n) \pi(u | y) Q_u^{(q)}(m | n).$$

The forward final ensemble is

$$p_{k+1}(m) = \sum_{n,y,u} P_F^{(k,q)}(n, y, u, m),$$

and the sensor marginal is

$$p_{Y,k}(y) = \sum_n p_k(n) S(y | n).$$

As in the revised $q = 1$ theory, use the finite-time reverse reference

$$P_R^{(k,q)}(m, y, u, n) = p_{k+1}(m) p_{Y,k}(y) \pi(u | y) Q_u^{(q)}(n | m).$$

The reverse process starts from the forward final ensemble. There is no requirement that $p_{k+1} = p_k$.

11 Finite-time path irreversibility and second-law identity

Define the path log-ratio

$$\Sigma_{k,q}^* = \ln \frac{P_F^{(k,q)}(n, y, u, m)}{P_R^{(k,q)}(m, y, u, n)}.$$

The policy factor cancels. Introduce pointwise sensing information

$$i_k(n; y) = \ln \frac{S(y | n)}{p_{Y,k}(y)},$$

stochastic system entropy

$$s_{\text{sys}}(n; p_k) = -\ln p_k(n) + S_{\text{mix}}(n),$$

and controlled current increment

$$j_c = u(m - n).$$

Using the transition-ratio theorem gives the pathwise identity

$$\boxed{\Sigma_{k,q}^* = \Delta s_{\text{sys}} + h j_c + i_k(n; y).}$$

Averaging over forward paths yields

$$\boxed{\Delta S_{\text{sys},k} + h J_{c,q,k} + I(n_k; Y_k) = \Sigma_{k,q} = D_{\text{KL}}(P_F^{(k,q)} \| P_R^{(k,q)}) \geq 0.}$$

The system entropy is

$$S_{\text{sys}}[p] = -\sum_n p(n) \ln p(n) + \sum_n p(n) \ln \binom{N}{n}.$$

This is a finite-time identity. The boundary/storage term ΔS_{sys} is retained explicitly, so no equilibrium or stationarity assumption is required.

12 Thermodynamic control efficiency for general q

Remove the transient storage contribution from path irreversibility:

$$\boxed{C_{\text{th},q,k} = \Sigma_{k,q} - \Delta S_{\text{sys},k} = h J_{c,q,k} + I(n_k; Y_k).}$$

For target-directed operation $h J_{c,q,k} \geq 0$ and $C_{\text{th},q,k} > 0$, define

$$\boxed{\eta_{\text{th},q,k} = \frac{h J_{c,q,k}}{h J_{c,q,k} + I(n_k; Y_k)} \leq 1.}$$

No literal energetic work interpretation is needed: the numerator is the affinity-weighted directed population current and the denominator is the non-storage control expenditure appearing in the exact path identity.

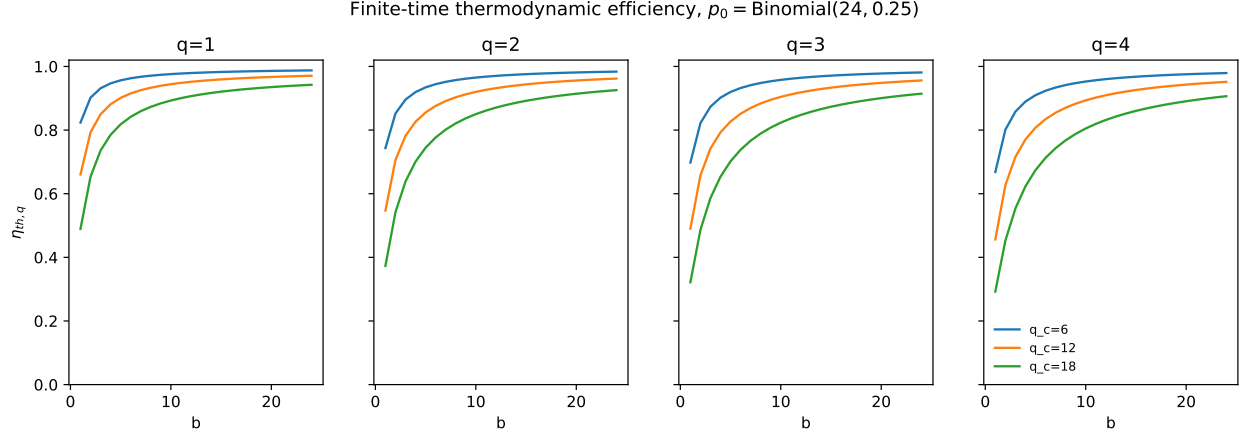


Figure 11: Finite-time thermodynamic control efficiency for $p_0(n) = \text{Binomial}(24, 0.25)$, $h = 2$, $\gamma = 0.75$, $\beta = 4$, $\theta = 1/2$, and the canonical gate. Each panel varies b for three sensing budgets. The $q = 1$ panel numerically reproduces the previous report; at fixed resources, the illustrative $q > 1$ kinetics reduce current and therefore lower efficiency.

At the old checkpoint $b = 18$, the exact $q = 1$ values are

$$\eta_{\text{th}} = 0.9848618, 0.9644058, 0.9303014 \quad \text{for } q_c = 6, 12, 18,$$

matching the previous report to machine precision.

12.1 Current–irreversibility view

The efficiency ratio does not replace the direct geometry of

$$(J_{c,q}, \Sigma_q).$$

A control-design problem can still be posed as minimizing irreversibility for a required current or maximizing current under an irreversibility budget.

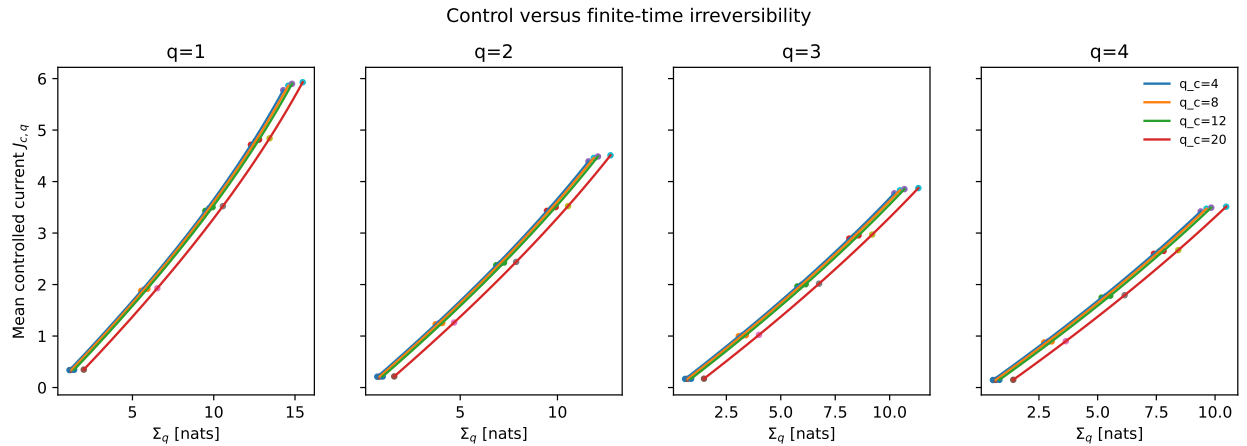


Figure 12: Generalized current versus finite-time path irreversibility. Each curve varies $b = 1, \dots, 24$; panels vary q , and curves within a panel vary q_c . Residual social context changes the attainable current–irreversibility geometry even though the affinity term in the path ratio is unchanged.

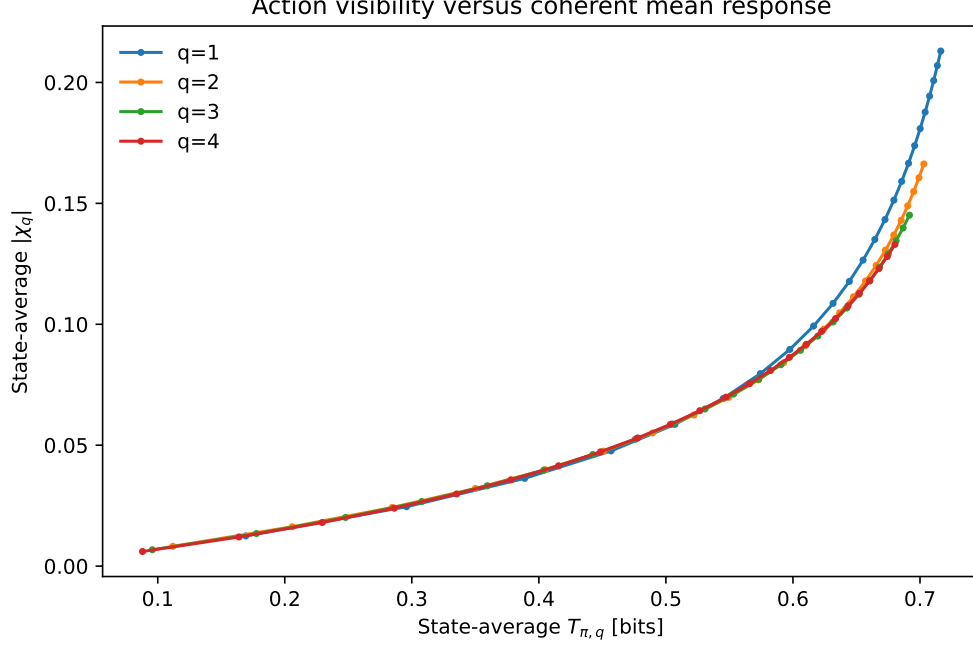


Figure 13: Action visibility versus coherent mean response. Under the illustrative closure, increasing q can reduce $|\chi_q|$ more strongly than it reduces $T_{\pi,q}$, illustrating why action-dependent distributional change and directed mean control should remain separate observables.

13 Finite-horizon aggregation

The one-cycle quantities generally vary with k . Summing the path identity over H cycles gives

$$S_{\text{sys}}[p_H] - S_{\text{sys}}[p_0] + h \sum_{k=0}^{H-1} J_{c,q,k} + \sum_{k=0}^{H-1} I(n_k; Y_k) = \sum_{k=0}^{H-1} \Sigma_{k,q}.$$

The finite-horizon non-storage expenditure is

$$C_{\text{th},q}^{(H)} = h \sum_k J_{c,q,k} + \sum_k I(n_k; Y_k),$$

and the corresponding efficiency is the ratio of accumulated terms,

$$\eta_{\text{th},q}^{(H)} = \frac{h \sum_k J_{c,q,k}}{h \sum_k J_{c,q,k} + \sum_k I(n_k; Y_k)}.$$

It is not the arithmetic mean of the cycle-wise efficiencies.

14 Operational calibration for $q \geq 1$

The generalized theory becomes empirical through controlled microscopic transitions conditioned on residual peer context. At fixed (q, j) estimate

$$p_+^{(q,j)} = P(\text{non-Z} \rightarrow Z \mid \text{controlled}, q, J = j),$$

$$p_-^{(q,j)} = P(Z \rightarrow \text{non-Z} \mid \text{controlled}, q, J = j).$$

Then

$$\gamma_{q,j}^{\text{eff}} = p_+^{(q,j)} + p_-^{(q,j)}, \quad h_{q,j}^{\text{eff}} = \ln \frac{p_+^{(q,j)}}{p_-^{(q,j)}}.$$

The primary single-affinity hypothesis is empirically testable:

$$h_{q,j}^{\text{eff}} \approx h \quad \text{across supported contexts,}$$

while $\gamma_{q,j}^{\text{eff}}$ is allowed to vary strongly with j and q . The preferred empirical workflow is therefore to estimate one common h , estimate the context compliances $\gamma_q(j)$, build $G_q(n)$ from the exact hypergeometric law, and then compute K_q , χ_q , $T_{\pi,q}$, $J_{c,q}$ and the finite-time thermodynamic quantities without fitting a separate population-level model.

The supplied Study 09i analysis archive contains micro-slot before/after focal opinions and identifies controlled slots, but its exported micro-slot table does not retain the residual peer vote composition J . Consequently, the context-conditioned $h_{q,j}^{\text{eff}}$ diagnostic cannot be reconstructed exactly from that archive alone. The implementation therefore provides the calibration API, but this report does not invent a fitted $\gamma_q(j)$ from insufficient records.

If future context-conditioned estimates show systematic variation in $h_{q,j}^{\text{eff}}$ beyond sampling uncertainty, the single-affinity hypothesis is falsified. In that case a state-dependent edge affinity $A_q(n)$ and discrete potential $\Psi_q(n)$ provide the correct fallback; this is intentionally separated from the primary model rather than silently absorbed into G_q .

15 $q = 1$ reduction and numerical validation

The new implementation was compared directly against the preserved `theory_revised.py` baseline over multiple parameter tuples and the full object chain: sensing, policy, microscopic kernel, Q_1 , susceptibility, T_π , Pinsker bound, η_{IR} , propagation, current, one-cycle thermodynamics, direct path KL, finite-horizon totals, and current moments.

The maximum differences were:

Object family	maximum absolute difference
Microscopic kernel K	0
$Q_1 = K^b$	0
Susceptibility	1.30×10^{-15}
T_π	0
η_{IR}	2.78×10^{-16}
One-cycle thermodynamic quantities	3.55×10^{-14}
Finite-horizon quantities	6.40×10^{-14}
Old-report figure data	1.24×10^{-14}

The maximum direct-KL identity residual over representative $q \in \{1, 2, 3, 4, 8, 24\}$ was 2.49×10^{-14} nats. Thus the generalized implementation reproduces the previous $q = 1$ theory to numerical precision and verifies the general- q path identity directly by finite sums.

16 What the new theory does and does not explain

The universal result is structural: if the $q-1$ residual peers modulate the controlled channel through an edge-symmetric kinetic factor, then q may substantially change response and information while the same directional affinity h remains sufficient for the finite-time stochastic-thermodynamic identity.

The behavioral shape of χ_q is *not* universal. A constant activity produces no q dependence. The canonical corroboration gate yields moderate low-state suppression and, for the displayed parameters, an increasingly interior response maximum for $q \geq 3$. A residual-unanimity gate produces a much stronger q -voter-like susceptibility window already at $q = 2$. This separation is desirable: the thermodynamic theorem is not tied to an arbitrary social-response closure.

For the canonical figure baseline $(N, h, q_c, \beta, \theta, b, \gamma) = (24, 2, 12, 4, 1/2, 12, 0.75)$, the exact one-cycle summary is

q	zero-response x^*	$\max \chi_q$	state-avg $T_{\pi,q}$	$J_{c,q}$	$\eta_{\text{th},q}$
1	0.8808	0.2790	0.6552	3.5043	0.9517
2	0.8824	0.1621	0.6096	2.4267	0.9318
3	0.8830	0.1200	0.5731	2.0071	0.9187
4	0.8833	0.1052	0.5473	1.7840	0.9094
8	0.8837	0.0885	0.4987	1.4313	0.8896

These numbers are illustrative theory outputs, not empirical fits.

17 Scope and limitations

1. n_k is a coarse population coordinate. LLM transitions can depend on epistemic state, active facts, dialogue history, reasoning and persistence.
2. Residual peer context is compressed into J and then into the chosen kinetic profile $\gamma_q(j)$.
3. The single-affinity hypothesis is empirical and falsifiable, not guaranteed.
4. q can alter both local transition kinetics and the occupied ensemble in the actual LLM process. The present theory conditions on the current p_k rather than modeling persistence or epistemic evolution as an additional thermodynamic force.
5. The construction is finite-time and does not require equilibrium or stationarity.
6. This is a stochastic-thermodynamic reference for the controlled population channel, not a claim of literal energy, temperature or a complete thermodynamics of language-model agents.

18 Direct finite-sum computation recipe

For one cycle:

1. Build $S(y | n)$, $\pi(1 | y)$ and a_n .
2. Build the exact residual-peer kernel $H_q(j | n)$.
3. Specify or calibrate $\gamma_q(j)$ and compute $G_q(n)$.

4. Build K_q , then $Q_0 = I$ and $Q_1^{(q)} = K_q^b$.
5. Compute $\chi_q = (Q_1^{(q)} x) - x$ and the exact $T_{\pi,q}(n)$.
6. Compute $J_{c,q} = N \sum_n p_k(n) a_n \chi_q(n/N)$.
7. Compute $I(n_k; Y_k)$ from p_k and the known sensor channel.
8. Propagate $p_{k+1} = p_k[(1 - a_n)Q_0 + a_n Q_1^{(q)}]$ row by row.
9. Compute ΔS_{sys} and evaluate $\Sigma_q = \Delta S_{\text{sys}} + hJ_{c,q} + I(n; Y)$.
10. Independently construct $P_F^{(q)}$ and $P_R^{(q)}$ and verify $\Sigma_q = D_{\text{KL}}(P_F^{(q)} \| P_R^{(q)})$.
11. For target-directed operation, compute $\eta_{\text{th},q} = hJ_{c,q}/[hJ_{c,q} + I(n; Y)]$.

No Monte Carlo approximation is required anywhere in the reference calculation.

Bottom line

The $q \geq 1$ extension can be obtained without adding a second directional force. The mathematically clean object is the edge activity $G_q(n)$ induced by the $q - 1$ residual peers. When that activity multiplies both directions of each population edge, it changes kinetics but cancels from the directional ratio. The full finite-time path identity and thermodynamic efficiency therefore survive unchanged in form. What remains empirical is the kinetic-context profile itself. That is exactly where the next comparison with the LLM data should be concentrated.